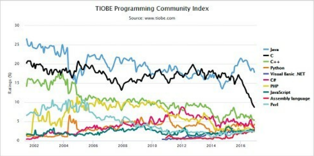

Figure 1: TIOBE Programming Community Index

the number of searches for a tutorial of a particular programming language. The assumption is that the more tutorials searched for, the more popular the language is. In this ranking also, languages preferred by academia might have an advantage over languages preferred by industry.

The IEEE Spectrum Ranking of Programming Languages has the most complicated ranking metric, which considers 12 factors from 10 sources. The 10 sources are Google Search, Google Trends, Twitter, GitHub, Stack Overflow, Reddit, Hacker News, CareerBuilder, Dice and IEEE Xplore Digital Library. Even though the IEEE Spectrum Ranking of Programming Languages looks like the most comprehensive ranking mechanism, the most popular and oft-quoted ranking scheme is TIOBE Index. Figure 1, taken from *www.tiobe.com*, shows the long-term TIOBE Index rankings of various programming languages updated up to December 2016.

Since each ranking scheme has a bias towards certain types of programming languages, selection based on a single ranking scheme will also be biased. So, for selecting languages I have considered all the four rankings, giving each equal weightage. Seven of the top 10 languages in all the four lists are the same. They are Java, C, C++, Python, C#, PHP, and JavaScript and so these languages have been included in my final list without much deliberation. The other three languages selected are Ruby, Go and Swift. They are in the top 20 of all the four ranking schemes. Ruby has two Top 10 finishes, whereas Go and Swift have one each. An additional selection criteria for Go and Swift is that both are gaining popularity and might be a permanent fixture in the top 10 list of popular programming languages in the near future, if not next year itself. So, the final list features Java, C, C++, Python, C#, PHP, JavaScript, Ruby, Go and Swift.

Since this article is appearing in a magazine called *Open Source For You*, the programming languages selected must have an open source angle. So, it is time to check whether these languages have any relevance to the open source development community by defining open source programming languages. Software, of course, can be open source or proprietary but what about a programming language? After all, a programming language is just a set of standards and specifications. A cursory search on the Internet tells you that nobody has really bothered to define an open source programming language. Many have listed open source programming languages without mentioning the eligibility criteria. So, I had to device a mechanism to check whether a language is open source or not. I have used a very relaxed criterion for qualification as an open source language -- if there is a compiler or interpreter with an open source licence existing for a programming language, then that programming language is deemed open source. Even C# and Swift qualify as open source languages based on my definition of open source languages, which is not a strict definition of course, but more a starting point for discussion.

Since my definition of open source languages is very liberal and inclusive, I am further proposing a metric called the Openness Score for the list. This depends on the impact of a programming language on open source project development. For example, a language like Swift, which is not widely used to develop open source software will have a low Openness Score, whereas C will have a high Openness Score because it is used to produce a lot of open source software. If you really want to contribute to the open source community, then learn one of the languages with a high Openness Score. Before moving on to the next topic, I would like to express my personal opinion about the term ‘open source programming language’. I believe the term ‘open standard’ describes the status of programming languages better than the term ‘open source’.

Now let’s continue with a discussion of the ten programming languages selected. The most important aspect regarding this discussion is analysing whether the popularity of a language is rising or falling. This definitely will help a lot of people to make educated guesses about which language to study next in order to make a difference in their career. Also, remember that the languages are being discussed in alphabetical order because it is not possible to rank them any further among themselves. This is because each of these programming languages is ranked differently in different ranking schemes.

C

C is a very popular programming language used for developing mobile, enterprise and embedded applications. It is a compiled language, which follows the imperative programming paradigm. The epic growth of C and UNIX is written on the same pages of history. C and Linux also have a very tightly knit relationship. Both are so connected that if you have to designate just a single programming language as open source, then there is no doubt that C is that language. The sad news regarding C is that in two of the rankings, it ranks low and in case of the TIOBE Index, though placed second, it had a huge drop in the ratings in 2016. But the IEEE Spectrum Ranking has placed C at the top. So, it is